LAB-6

SOFTWARE ENGINEERING

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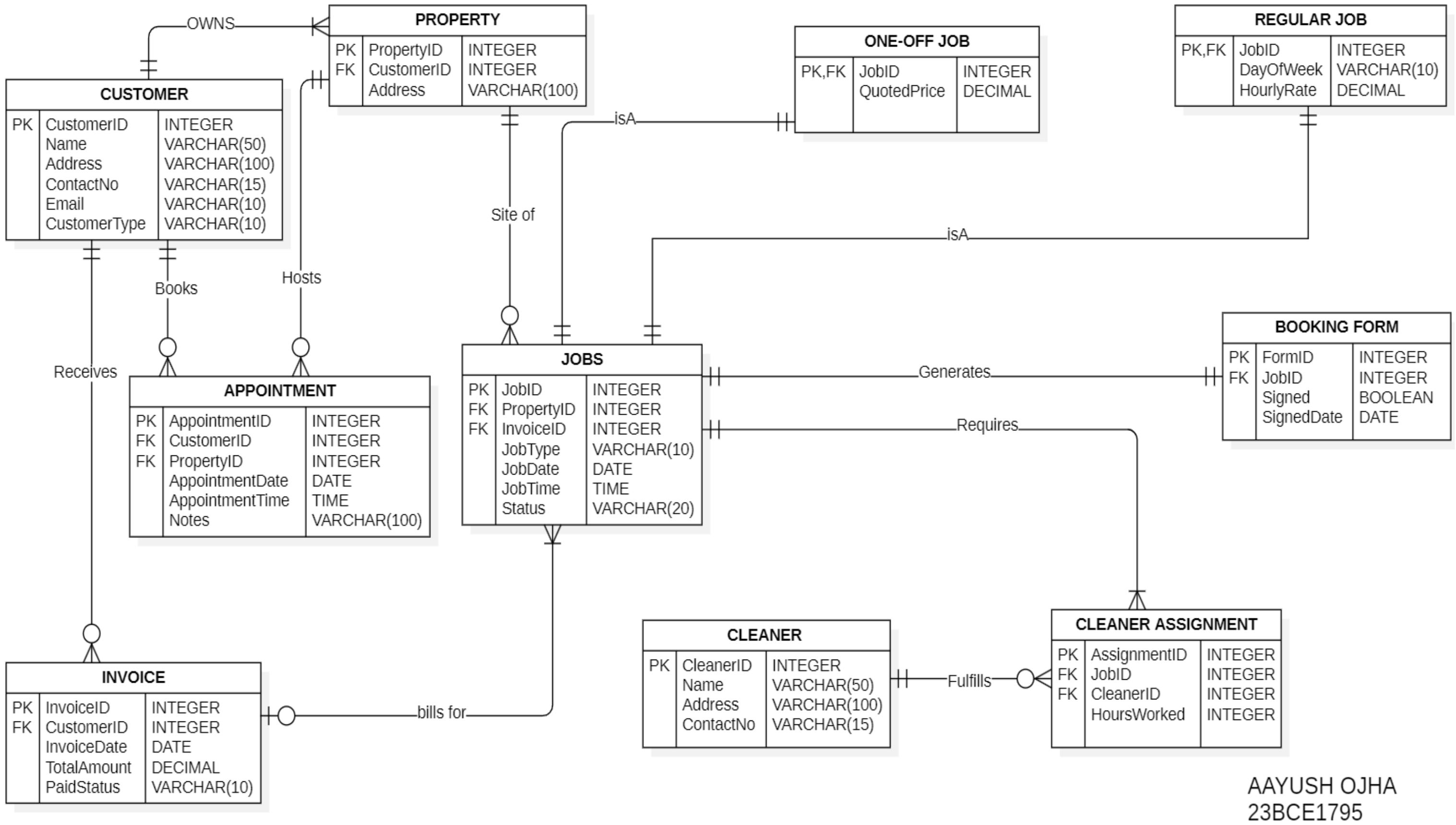
AIM-To develop ERD for JTJ and My Project.

JTJ ERD



EXPLAINATIONS-

1. CUSTOMER – PROPERTY

- Relationship: Owns
- Cardinality: One to Many
- Customer Side: Mandatory One (Every property belongs to exactly one customer)
- Property Side: Mandatory Many (Every customer must own at least one property)

2. CUSTOMER – APPOINTMENT

- Relationship: Books
- Cardinality: One to Many
- Customer Side: Mandatory One (Every appointment is booked by exactly one customer)
- Appointment Side: Optional Many (A customer can have zero (new customer) or many appointments)

3. PROPERTY – APPOINTMENT

- Relationship: Hosts
- Cardinality: One to Many
- Property Side: Mandatory One (Every appointment occurs at exactly one property)
- Appointment Side: Optional Many (A property can have zero or many appointments)

4. PROPERTY – JOB

- Relationship: Site Of
- Cardinality: One to Many
- Property Side: Mandatory One (Every job is located at exactly one property)
- Job Side: Optional Many (A property can have zero or many jobs recorded)

5. JOB – ONE OFF JOB

- Relationship: Is A (Specialisation)
- Cardinality: One to One
- Job Side: Mandatory One (The parent job record must exist)
- One Off Job Side: Mandatory One (Every One Off Job must correspond to exactly one Job record)

6. JOB – REGULAR JOB

- Relationship: Is A (Specialisation)
- Cardinality: One to One
- Job Side: Mandatory One (The parent job record must exist)
- Regular Job Side: Mandatory One (Every Regular Job must correspond to exactly one Job record)

7. JOB – CLEANER ASSIGNMENT

- Relationship: Requires
- Cardinality: One to Many
- Job Side: Mandatory One (Every assignment links to exactly one job)
- Cleaner Assignment Side: Mandatory Many (Every job must have at least one cleaner assigned)

8. CLEANER – CLEANER ASSIGNMENT

- Relationship: Fulfills
- Cardinality: One to Many
- Cleaner Side: Mandatory One (Every assignment belongs to exactly one cleaner)
- Cleaner Assignment Side: Optional Many (A cleaner can have zero or many active assignments)

9. JOB – BOOKING FORM

- Relationship: Generates
- Cardinality: One to One
- Job Side: Mandatory One (Every booking form corresponds to exactly one job)
- Booking Form Side: Mandatory One (Every job must generate exactly one booking form)

10. CUSTOMER – INVOICE

- Relationship: Receives
- Cardinality: One to Many
- Customer Side: Mandatory One (Every invoice is issued to exactly one customer)
- Invoice Side: Optional Many (A customer can receive zero or many invoices)

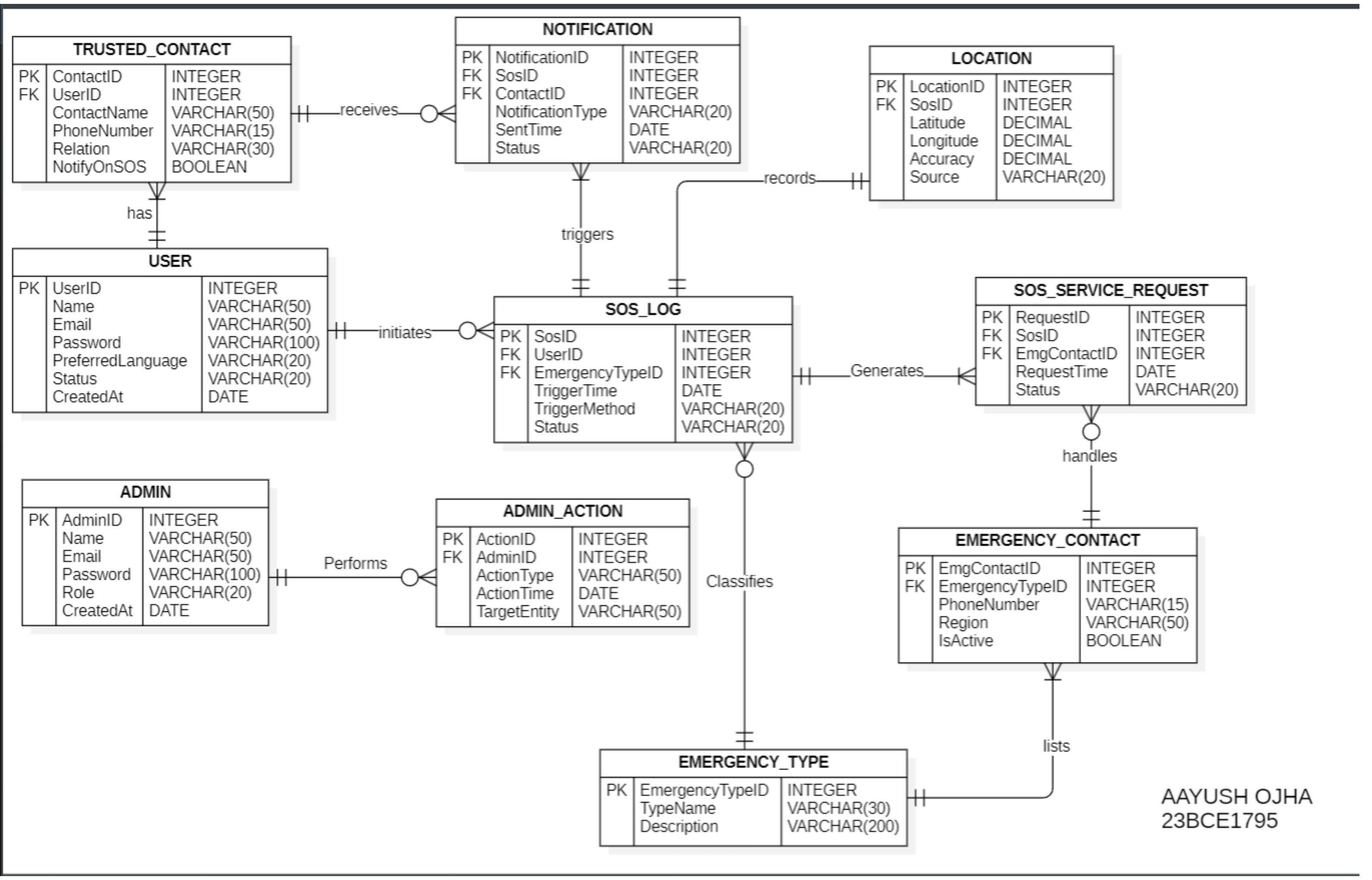
11. INVOICE – JOB

- Relationship: Bills For
- Cardinality: One to Many
- Invoice Side: Optional One (A job can exist without an invoice initially, but has max one invoice)
- Job Side: Mandatory Many (An invoice must contain at least one job to be created)

PROJECT ERD ON NEXT PAGE------------------------

PROJECT -Emergency Contact & SOS Awareness System

ERD



1. USER – TRUSTED_CONTACT

- Relationship: Has
- Cardinality: One to Many
- User Side: Mandatory One (Every trusted contact belongs to exactly one user)
- Trusted Contact Side: Mandatory Many (Every user must have at least one trusted contact registered)

2. USER – SOS_LOG

- Relationship: Initiates
- Cardinality: One to Many

- User Side: Mandatory One (Every SOS log is initiated by exactly one user)
- SOS Log Side: Optional Many (A user can trigger zero or many SOS alerts)

3. SOS_LOG – NOTIFICATION

- Relationship: Triggers
- Cardinality: One to Many
- SOS Log Side: Mandatory One (Every notification is linked to exactly one SOS event)
- Notification Side: Mandatory Many (Every SOS event must trigger at least one notification)

4. TRUSTED_CONTACT – NOTIFICATION

- Relationship: Receives
- Cardinality: One to Many
- Trusted Contact Side: Mandatory One (Every notification record identifies exactly one specific recipient)
- Notification Side: Optional Many (A trusted contact can receive zero or many notifications)

  A single SOS event generates multiple notification records—one for each contact selected.

5. SOS_LOG – LOCATION

- Relationship: Records
- Cardinality: One to One
- SOS Log Side: Mandatory One (Every location record belongs to exactly one SOS log)
- Location Side: Mandatory One (Every SOS log must record exactly one location snapshot)

6. SOS_LOG – SOS_SERVICE_REQUEST

- Relationship: Generates
- Cardinality: One to Many
- SOS Log Side: Mandatory One (Every request is generated by exactly one SOS event)
- Service Request Side: Mandatory Many (Every SOS event must generate at least one service request)

7. EMERGENCY_CONTACT – SOS_SERVICE_REQUEST

- Relationship: Handles

- Cardinality: One to Many
- Emergency Contact Side: Mandatory One (Every request is handled by exactly one emergency service contact)
- Service Request Side: Optional Many (An emergency contact can handle zero or many requests)

8. EMERGENCY_TYPE – SOS_LOG

- Relationship: Classifies
- Cardinality: One to Many
- Emergency Type Side: Mandatory One (Every SOS log must have exactly one type, e.g., Medical)
- SOS Log Side: Optional Many (An emergency type can be associated with zero or many logs)

9. EMERGENCY_TYPE – EMERGENCY_CONTACT

- Relationship: Lists
- Cardinality: One to Many
- Emergency Type Side: Mandatory One (Every contact belongs to exactly one category type)
- Emergency Contact Side: Mandatory Many (Every emergency type must have at least one contact number listed)

10. ADMIN – ADMIN_ACTION

- Relationship: Performs
- Cardinality: One to Many
- Admin Side: Mandatory One (Every action is performed by exactly one admin)
- Admin Action Side: Optional Many (An admin can perform zero or many audit actions)

CONCLUSION-

Hence, ERD was created for both JTJ and My project. The Entity-Relationship Diagram (ERD) follows standard Crow's Foot Notation, where vertical bars (|) denote mandatory existence and circles (O) denote optional existence in the relationship constraints.